管理及監控BranchCache組態

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管理及監控BranchCache組態

修改BranchCache組態

您可以修改SVM上的BranchCache服務組態、包括變更雜湊存放區目錄路徑、雜湊存放區最大目錄大小、作業模式、以及支援哪些版本的BranchCache。您也可以增加包含雜湊存放區的磁碟區大小。

步驟

1. 執行適當的行動：

如果您想要...	輸入下列項目...
修改雜湊存放區目錄大小	「Vserver CIFS BranchCache modify -vserver vserver_name -hash-store-max-size｛integer[kb
MB	GB
TB	PB]｝」
增加包含雜湊存放區的磁碟區大小	「Volume size -vserver vserver_name -volume volume volume volume名稱-new-size new_size[k
	g
t]」如果包含雜湊存放區的磁碟區已滿、您可能可以增加磁碟區的大小。您可以將新的Volume大小指定為數字、然後再指定單位。 請參閱 "邏輯儲存管理" 以取得有關增加磁碟區大小的詳細資訊。	修改雜湊存放區目錄路徑
「Vserver CIFS BranchCache modify -vserver vserver_name -hash-store-path -flush-hashes｛true	fals｝」如果SVM是SVM災難恢復來源、則雜湊路徑無法位於根磁碟區上。這是因為根磁碟區並未複寫到災難恢復目的地。 BranchCache雜湊路徑可以包含空白和任何有效的檔案名稱字元。 如果您修改雜湊路徑、則必須使用「-Flush雜湊」參數、指定ONTAP 是否要讓效益分析從原始雜湊存放區位置清除雜湊。您可以為「齊面雜湊」參數設定下列值： 如果您指定「**true**」、**ONTAP** 則當啟用了**BranchCache**的用戶端發出新要求時、會刪除原始位置的雜湊、並在新位置建立新的雜湊。 如果您指定「假」、則不會清除雜湊。 + 在這種情況下、您可以選擇稍後再重複使用現有的雜湊、方法是將雜湊存放區路徑變更回原始位置。

如果您想要...	輸入下列項目...
變更操作模式	「Vserver CIFS BranchCache modify -vserver vserver_name -operation-tMode｛per共享區
all-shares	停用｝」 [NOTE] ==== 修改操作模式時、您應該注意下列事項：- ONTAP 當SMB工作階段設定完成時、就會向您宣傳支援某個共用區的BranchCache。-啟用了BranchCache之後、已建立工作階段的用戶端必須中斷連線並重新連線、才能使用此共用區的快取內容。 ====
變更支援的BranchCache版本	「Vserver CIFS BranchCache modify -vserver vserver_name -versions｛v1-enable
v2-enable	enable All｝」

2. 使用「vserver CIFS BranchCache show」命令來驗證組態變更。

"邏輯儲存管理指南"

顯示有關BranchCache組態的資訊

您可以在儲存虛擬機器（SVM）上顯示有關BranchCache組態的資訊、這些資訊可在驗證組態或在修改組態之前判斷目前設定時使用。

步驟

1. 執行下列其中一項動作：

如果您要顯示...	輸入此命令...
關於所有SVM上的BranchCache組態的摘要資訊	「Vserver CIFS BranchCache show」
有關特定SVM組態的詳細資訊	「Vserver CIFS BranchCache show -vserver *vserver_name*」

下列範例顯示SVM VS1上的BranchCache組態相關資訊：

```
cluster1::> vserver cifs branchcache show -vserver vs1

                                   Vserver: vs1
            Supported BranchCache Versions: enable_all
                        Path to Hash Store: /hash_data
            Maximum Size of the Hash Store: 20GB
  Encryption Key Used to Secure the Hashes: -
         CIFS BranchCache Operating Modes: per_share
```

變更BranchCache伺服器金鑰

您可以修改儲存虛擬機器（SVM）上的BranchCache組態、並指定不同的伺服器金鑰、藉此變更BranchCache伺服器金鑰。

您可以將伺服器金鑰設定為特定值、以便在多個伺服器為相同檔案提供BranchCache資料時、用戶端可以使用相同伺服器金鑰來自任何伺服器的雜湊。

變更伺服器金鑰時、您也必須清除雜湊快取。在清除雜湊之後、ONTAP 當啟用了BranchCache的用戶端發出新要求時、會建立新的雜湊。

步驟

1. 使用下列命令變更伺服器金鑰：「vserver CIFS BranchCache modify -vserver vserver_name -server_key text -flush-hahes true」

   設定新的伺服器金鑰時、您也必須指定「-Flush雜湊」、並將值設為「true」。

2. 使用「vserver CIFS BranchCache show」命令來驗證BranchCache組態是否正確。

下列範例設定新的伺服器金鑰、其中包含空格並清除SVM VS1上的雜湊快取：

```
cluster1::> vserver cifs branchcache modify -vserver vs1 -server-key "new
vserver secret" -flush-hashes true

cluster1::> vserver cifs branchcache show -vserver vs1

                                   Vserver: vs1
            Supported BranchCache Versions: enable_all
                        Path to Hash Store: /hash_data
            Maximum Size of the Hash Store: 20GB
  Encryption Key Used to Secure the Hashes: -
         CIFS BranchCache Operating Modes: per_share
```

理由ONTAP 不驗證BranchCache雜湊

在指定路徑上預先計算的BranchCache雜湊

您可以將BranchCache服務設定為預先計算單一檔案、目錄或目錄結構中所有檔案的雜湊。如果您想要在關閉非尖峰時段內、針對啟用了BranchCache的共用區中的資料運算雜湊、這項功能就很有幫助。

如果您想在顯示雜湊統計資料之前先收集資料樣本、則必須使用「統計資料開始」和選用的「統計資料停止」命令。

- 您必須指定要預先計算雜湊的儲存虛擬機器（SVM）和路徑。
- 您也必須指定是否要以循環方式計算雜湊。
- 如果您想要以遞迴方式計算雜湊、則BranchCache服務會在指定路徑下遍歷整個目錄樹狀結構、並針對每個符合條件的物件來計算雜湊。

步驟

1. 視需要預先運算雜湊：

如果您想要預先計算雜湊...	輸入命令...
單一檔案或目錄	「Vserver CIFS BranchCache hash-create -vserver vserver_name -path path -recurse假」
以循環方式處理目錄結構中的所有檔案	「Vserver CIFS BranchCache hash-create -vserver vserver_name -path falure_path -recurse true」

2. 使用「統計資料」命令來驗證雜湊是否正在計算：
   a. 顯示所需SVM執行個體上「hashd」物件的統計資料：「統計資料show -object hashd -instance vserver_name」
   b. 重複執行命令、確認所建立的雜湊數量增加。

以下範例會在路徑「/data」上建立雜湊、並在SVM VS1上的所有內含檔案和子目錄上建立雜湊：

```
cluster1::> vserver cifs branchcache hash-create -vserver vs1 -path /data
-recurse true

cluster1::> statistics show -object hashd -instance vs1
Object: hashd
Instance: vs1
Start-time: 9/6/2012 19:09:54
End-time: 9/6/2012 19:11:15
Cluster: cluster1

    Counter                                                     Value
    -------------------------------- --------------------------------
    branchcache_hash_created                                       85
    branchcache_hash_files_replaced                                 0
    branchcache_hash_rejected                                       0
    branchcache_hash_store_bytes                                    0
    branchcache_hash_store_size                                     0
    instance_name                                                 vs1
    node_name                                                   node1
    node_uuid
                                11111111-1111-1111-1111-111111111111
    process_name                                                    -

cluster1::> statistics show -object hashd -instance vs1
Object: hashd
Instance: vs1
Start-time: 9/6/2012 19:09:54
End-time: 9/6/2012 19:11:15
Cluster: cluster1

    Counter                                                     Value
    -------------------------------- --------------------------------
    branchcache_hash_created                                       92
    branchcache_hash_files_replaced                                 0
    branchcache_hash_rejected                                       0
    branchcache_hash_store_bytes                                    0
    branchcache_hash_store_size                                     0
    instance_name                                                 vs1
    node_name                                                   node1
    node_uuid
                                11111111-1111-1111-1111-111111111111
    process_name                                                    -
```

"效能監控設定"

從SVM BranchCache雜湊存放區清除雜湊

您可以清除儲存虛擬機器（SVM）上的BranchCache雜湊存放區中的所有快取雜湊。如果您已變更分公司的BranchCache組態、這項功能就很有用。例如、如果您最近將快取模式從分散式快取重新設定為託管式快取模式、就會想要清除雜湊存放區。

在清除雜湊之後、ONTAP 當啟用了BranchCache的用戶端發出新要求時、會建立新的雜湊。

步驟

1. 從BranchCache雜湊存放區清除雜湊：「vserver CIFS BranchCache hash-flush -vserver *vserver_name*」

   「Vserver CIFS BranchCache hash-flush -vserver VS1」

顯示BranchCache統計資料

您可以顯示BranchCache統計資料、以識別快取效能的表現、判斷您的組態是否提供快取內容給用戶端、並判斷是否刪除雜湊檔案、以留出空間來儲存較新的雜湊資料。

「hashd」統計物件包含提供有關BranchCache雜湊統計資訊的計數器。「CIFS」統計物件包含提供有關BranchCache相關活動統計資訊的計數器。您可以在進階權限層級收集及顯示這些物件的相關資訊。

步驟

1. 將權限等級設為進階：「et -priv榮幸advanc進 階」

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them
only when directed to do so by support personnel.
Do you want to continue? {y|n}: y
```

2. 使用「統計資料目錄計數器show」命令來顯示與BranchCache相關的計數器。

   如需統計資料計數器的詳細資訊、請參閱此命令的手冊頁。

```
cluster1::*> statistics catalog counter show -object hashd

Object: hashd
    Counter                     Description
    ---------------------------
----------------------------------------------
    branchcache_hash_created    Number of times a request to generate
                                BranchCache hash for a file succeeded.
    branchcache_hash_files_replaced
                                Number of times a BranchCache hash file
was
```

```
                                deleted to make room for more recent
hash
                                data. This happens if the hash store
size is
                                exceeded.
    branchcache_hash_rejected   Number of times a request to generate
                                BranchCache hash data failed.
    branchcache_hash_store_bytes
                                Total number of bytes used to store hash
data.
    branchcache_hash_store_size Total space used to store BranchCache
hash
                                data for the Vserver.
    instance_name               Instance Name
    instance_uuid               Instance UUID
    node_name                   System node name
    node_uuid                   System node id
9 entries were displayed.

cluster1::*> statistics catalog counter show -object cifs

Object: cifs
    Counter                     Description
    ---------------------------
----------------------------------------------
    active_searches             Number of active searches over SMB and
SMB2
    auth_reject_too_many        Authentication refused after too many
                                requests were made in rapid succession
    avg_directory_depth         Average number of directories crossed by
SMB
                                and SMB2 path-based commands
    avg_junction_depth          Average number of junctions crossed by
SMB
                                and SMB2 path-based commands
    branchcache_hash_fetch_fail Total number of times a request to fetch
hash
                                data failed. These are failures when
                                attempting to read existing hash data.
It
                                does not include attempts to fetch hash
data
                                that has not yet been generated.
    branchcache_hash_fetch_ok   Total number of times a request to fetch
hash
                                data succeeded.
```

```
    branchcache_hash_sent_bytes Total number of bytes sent to clients
                                requesting hashes.
    branchcache_missing_hash_bytes
                                Total number of bytes of data that had
to be
                                read by the client because the hash for
that
                                content was not available on the server.
   ....Output truncated....
```

3. 使用「統計資料開始」和「統計資料停止」命令來收集與BranchCache相關的統計資料。

```
cluster1::*> statistics start -object cifs -vserver vs1 -sample-id 11
Statistics collection is being started for Sample-id: 11

cluster1::*> statistics stop -sample-id 11
Statistics collection is being stopped for Sample-id: 11
```

4. 使用「統計資料顯示」命令顯示所收集的BranchCache統計資料。

```
cluster1::*> statistics show -object cifs -counter
branchcache_hash_sent_bytes -sample-id 11

Object: cifs
Instance: vs1
Start-time: 12/26/2012 19:50:24
End-time: 12/26/2012 19:51:01
Cluster: cluster1

    Counter                                                     Value
    -------------------------------- --------------------------------
    branchcache_hash_sent_bytes                                     0
    branchcache_hash_sent_bytes                                     0
    branchcache_hash_sent_bytes                                     0
    branchcache_hash_sent_bytes                                     0

cluster1::*> statistics show -object cifs -counter
branchcache_missing_hash_bytes -sample-id 11

Object: cifs
Instance: vs1
Start-time: 12/26/2012 19:50:24
End-time: 12/26/2012 19:51:01
Cluster: cluster1

    Counter                                                     Value
    -------------------------------- --------------------------------
    branchcache_missing_hash_bytes                                  0
    branchcache_missing_hash_bytes                                  0
    branchcache_missing_hash_bytes                                  0
    branchcache_missing_hash_bytes                                  0
```

5. 返回管理員權限等級：「et -priv. admin」

```
cluster1::*> set -privilege admin
```

顯示統計資料

"效能監控設定"

支援BranchCache群組原則物件

支援BranchCache群組原則物件（GPO）、可集中管理某些BranchCache組態參

數。ONTAP有兩個GPO用於BranchCache：「BranchCache GPO的雜湊發佈」和「BranchCache GPO的雜湊版本支援」。

- *《BranchCache GPO*雜湊》發佈

  BranchCache GPO的雜湊發佈對應於「作業模式」參數。當發生GPO更新時、此值會套用至群組原則所套用之組織單位（OU）內的儲存虛擬機器（SVM）物件。

- *支援BranchCache GPO *的雜湊版本

  支援的雜湊版本對應於「-versions」參數。當發生GPO更新時、此值會套用至群組原則套用至的組織單位內所包含的SVM物件。

將群組原則物件套用至CIFS伺服器

顯示有關BranchCache群組原則物件的資訊

您可以顯示有關CIFS伺服器的群組原則物件（GPO）組態資訊、以判斷是否為CIFS伺服器所屬的網域定義了BranchCache GPO、如果是、允許的設定為何。您也可以決定是否將BranchCache GPO設定套用至CIFS伺服器。

即使在CIFS伺服器所屬的網域中定義了GPO設定、但不一定會套用至包含CIFS型儲存虛擬機器（SVM）的組織單位（OU）。套用的GPO設定是套用至CIFS型SVM的所有已定義GPO的子集。透過GPO套用的BranchCache設定會覆寫透過CLI套用的設定。

步驟

1. 使用「vserver CIFS group原則show-defined」命令、顯示Active Directory網域的已定義的BranchCache GPO設定。

   此範例不會顯示命令的所有可用輸出欄位。輸出被截短。

```
cluster1::> vserver cifs group-policy show-defined -vserver vs1

Vserver: vs1
-----------------------------
       GPO Name: Default Domain Policy
       Level: Domain
      Status: enabled
  Advanced Audit Settings:
      Object Access:
          Central Access Policy Staging: failure
  Registry Settings:
      Refresh Time Interval: 22
      Refresh Random Offset: 8
      Hash Publication Mode for BranchCache: per-share
      Hash Version Support for BranchCache: version1
  [...]

    GPO Name: Resultant Set of Policy
      Status: enabled
  Advanced Audit Settings:
      Object Access:
          Central Access Policy Staging: failure
  Registry Settings:
      Refresh Time Interval: 22
      Refresh Random Offset: 8
      Hash Publication for Mode BranchCache: per-share
      Hash Version Support for BranchCache: version1
  [...]
```

2. 使用「vserver CIFS group原則show-applied」命令、顯示套用至CIFS伺服器的「BranchCache GPO」設定。」

   此範例不會顯示命令的所有可用輸出欄位。輸出被截短。

```
cluster1::> vserver cifs group-policy show-applied -vserver vs1

Vserver: vs1
-----------------------------
     GPO Name: Default Domain Policy
       Level: Domain
      Status: enabled
  Advanced Audit Settings:
      Object Access:
          Central Access Policy Staging: failure
  Registry Settings:
      Refresh Time Interval: 22
      Refresh Random Offset: 8
      Hash Publication Mode for BranchCache: per-share
      Hash Version Support for BranchCache: version1
  [...]

    GPO Name: Resultant Set of Policy
       Level: RSOP
  Advanced Audit Settings:
      Object Access:
          Central Access Policy Staging: failure
  Registry Settings:
      Refresh Time Interval: 22
      Refresh Random Offset: 8
      Hash Publication Mode for BranchCache: per-share
      Hash Version Support for BranchCache: version1
 [...]
```

在CIFS伺服器上啟用或停用GPO支援

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